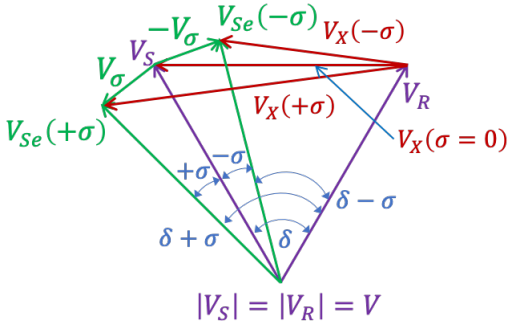
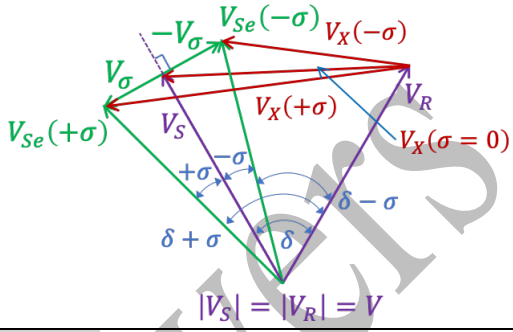
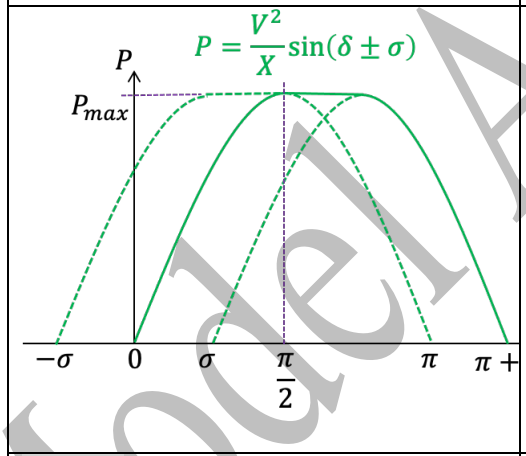
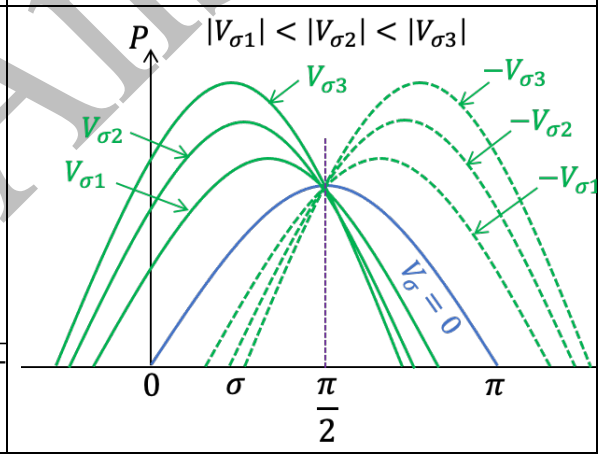
Question 1 on Flexible AC Transmission System (FACTS)

a) TCAPAR vs. QB

i) Phasor diagarm

Ideal phase angle compensation by TCPAR	Phase angle compensation by QB
	
[1 mark]	[1 mark]

ii) Power angle characteristics

Ideal phase angle compensation by TCPAR	Phase angle compensation by QB
	
[1 mark]	[1 mark]

iii) Injected voltage

Ideal phase angle compensation by TCPAR	Phase angle compensation by QB
The injected voltage (V_σ) is such that the effective voltage ($V_{se}=V_s+V_\sigma$) has the same magnitude as the original voltage (V_s) i.e. $\lvert V_{se}\rvert=\lvert V_s\rvert$	The injected voltage (V_σ) is in phase quadrature (leading or lagging) with respect to the original voltage (V_s) such that the effective voltage is always larger in magnitude than the original voltage i.e. $\lvert V_{se}\rvert=\sqrt{{V_s}^2+{V_\sigma}^2}>\lvert V_s\rvert$
[1 mark]	[1 mark]

Model Answers

iv) Components

Ideal phase angle compensation by TCPAR	Phase angle compensation by QB
Consists of power electronic converters with thyristor controlled elements alongside transformers.	Consists of a series and a shunt transformer with variable taps but no power electronic converter.
[1 mark]	[1 mark]

b) TCSC

i) $\frac{P'_{max}}{P_{max}} = \frac{X}{X(1-k)} = 2$

$k = 0.5$ or 50%

[2 marks]

ii) $P_{max0} = \frac{1}{X} = 3.33$ p.u.

[1 mark]

$P = 2.0$ p.u.

$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max0}}\right) = 36.87°$

[1 mark]

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 71.77°$

[2 marks]

iii) $k = 0.3$

$P_{max1} = \frac{1}{X(1-k)} = 4.762$ p.u.

[1 mark]

$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 24.835°$

[1 mark]

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 87.257°$

Model Answers

[1 mark]

iv) Under the condition specified in the problem, the critical clearing angle (δ_{cr}) depends only on the initial power angle (δ_0) which depend on $\frac{P}{P_{max}}$

[1 mark]

In part (ii)

$$\frac{P}{P_{max}} = \frac{2.0}{3.33} = 0.6$$

[1 mark]

If P' is the power transfer possible with the same critical clearing angle (δ_{cr}) as in part (ii)

$$\frac{P'}{P_{max1}} = 0.6 \rightarrow P' = 0.6 \times P_{max1} = 2.857 \text{ p.u.}$$

[1 mark]

Model Answers

Question 2 on System Operation with Low Inertia

a)

i)

Rotor angle stability
Frequency stability
Voltage stability
Resonance stability
Converter driven stability
System protection

[1 mark for One, 2 marks for Three, 3 marks for Five]

ii)

$2H\dot{f}(t) = R(t) - \Delta P_L,$
$R(t) = \frac{R}{T_d}t, \ \ if\ t < T_d,$
$\therefore f(t) = \frac{R}{4HT_d}t^2 - \frac{\Delta P_L}{2H}t,$
[1 mark]

$t_n = \frac{\Delta P_L T_d}{R},$
$f_{nadir} = |\frac{\Delta P_L^2 T_d}{4HR}|,$
[1 mark]

$\therefore HR \geq \frac{\Delta P_L^2 T_d}{4f_{max}}.$
[1 mark]

iii)

$$\left(H - \mathrm{DC}\times\frac{\mathrm{T_{DC}}}{4\times\Delta f_{\max}}\right)\times R \geq \frac{(P_{\mathrm{L}} - \mathrm{DC})^2 \times \mathrm{T_d}}{4\times\Delta f_{\max}}$$

When DC = 0.55 GW
$R = 1.9$ GW
[1 mark]
$g(R, EFR) = (3 - 0.2EFR)R - 3.5(1.8 - EFR)^2,$

[1 mark]

$\frac{\partial g}{\partial R} = 3 - 0.2EFR = 2.891,$
$\frac{\partial g}{\partial EFR} = -0.2R + 2 * 3.5(1.8 - EFR) = 8.4035,$
$\frac{\partial R}{\partial EFR} = \frac{\frac{\partial g}{\partial EFR}}{\frac{\partial g}{\partial R}} = 2.907,$
$\frac{\partial R}{\partial EFR} \times \frac{c_{EFR}}{c_R} = \frac{R}{EFR} = 3.5,$
$\frac{c_{EFR}}{c_R} = 1.204.$
[2 marks]

Model Answers

b)

i) BOOKWORK. Wind turbines control their power intake and their speed using two actuators: the torque of the generator, which is controlled using power electronic converter, and the blade pitch angle, which is controlled using a servomotor.
Headroom can be created by using either actuator. A greater blade pitch angle could be kept in place at all times to be able to increase the torque by reducing the angle at any time.
At low and medium speeds, the torque could be used to stabilise the speed at an operating point with a higher tip-speed ratio than the one that corresponds to the maximum power point. This would enable the turbine to boost the output power by increasing the torque, which would both extract kinetic energy from the turbine to deliver power to the grid and increase the power intake from the wind on the longer term by moving closer to the maximum power point.

[4 marks]

ii) The equation of the synthetic inertia is based on the swing equation:

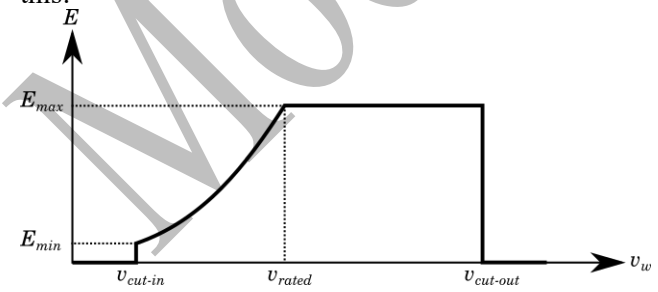
$$\Delta P = \frac{-2H}{f_{nominal}} S_{base} \frac{df}{dt}$$

This can be used to calculate how much the power output of the generator would change for a given RoCoF. A positive RoCoF would cause the power to reduce, which in terms of its implementation in a wind turbine it would imply a reduction of the power output, which can be achieved by using speed and blade pitch angle control. A negative RoCoF would cause the power to increase, which requires the generator and its power converter to be sized for this increment. In this problem we know that our converter can handle an increase of power output of 1 MW, we can use this figure to calculate the highest *H* it would be able to emulate:

$$H = -\frac{\Delta P \, f_{nominal}}{2 \, S_{base} \frac{df}{dt}} = -\frac{1\ \text{MW} \cdot 50\ \text{Hz}}{2 \cdot 8\ \text{MW} \cdot (\text{-}2\ \text{Hz/s})} = 1.56\ s$$

[3 marks]

iii) The turbine is at rest for wind speeds below cut-in or above cut-out. The turbine will spin at a speed that is linearly proportional to the wind speed under the Maximum Power Point Tracking (MPPT) region when the output power yield is below rated power. The turbine will operate at constant speed at any wind speed between rated speed and cut-out. The kinetic energy of the turbine is proportional to the square of the rotating speed, therefore the kinetic energy versus wind speed would look like this:



While it is not strictly necessary for candidates to explicitly say that the energy will increase with the square of the wind speed in the MPPT region, they are expected to produce a graph similar to the one shown below and to justify the approximate shape of the curve considering the behaviour of the turbine under the different ranges of wind speed.

[3 marks]

Model Answers

Question 3 on Power System State Estimation

a)

i) Why is state estimation necessary in power systems? [4]

The operator needs to know in the operation room what are the flows across various circuits, how much capacity head room are available for various assets, what are the voltages at different locations etc- these are necessary to operate the system without violating the limits of any asset. The answers to this query cannot come directly from measurements, as many of them are not directly measured. So, only through state estimation these are possible to be obtained.

ii) What is weighted least square method of state estimation? [3]

``*This is a Maximum likelihood estimation where the quality of the measurements contributes to the state estimation through weights – higher weights are used for more accurate measurements; lower weights are for in accurate measurements. The error of the measurement is assumed to be treated as normal gaussian.*

iii) What are the advantages and disadvantages of this method? [3]

The solution is well developed, the handling of noise is effective as they are filtered out because the estimator is unbiased, so both positive and negative errors have cancelling effect. There is a filtering property. In the presence of in grossly in accurate measurement, the estimator has property to accord that less importance. The estimator cannot handle bad data as that is squared in the objective function

b) In the context of power distribution system

i) Why is state estimation not a routine task? [2]

The measurements are not routinely and comprehensively available. The network parameters are also not consistent, the accurate information on the network asymmetry is not available. The controllability of the control centre is also not possible to exercise in the absence of power flow control at the distribution voltage where it is seen as a passive system.

ii) How are the measurements classified for state estimation? [3]

Real, pseudo and zero injection types. In many cases measurements are not available, so historical load survey data is used.

(iii) How are these measurements represented in state estimation objective function? [3]

Zero injection can be represented as constraints to the state estimation objective as they are 100% accurate. The real measurement will have high weight; pseudo measurement should have low weights. The overall problem is a nonlinear objective with constraints that invokes penalty function for solutions.

Model Answers

iv) Why is it not straight forward to include the data from smart meters for state estimation? [2]

Smart meter data from customer locations and secondary substations are of very high time resolutions. But they are only available to power distribution control centre not in real time. Usually once or twice a day or even if they are available every 30 mins, still they are of limited use. They can be useful for generation/demand forecasting but not for state estimation.

[20 marks]

Model Answers

Question on Integrated Energy Systems

Part I

Describe the challenges for the power system in deep electrification of energy demand and how this can be addressed?

[3]

- Substantial capital deployment for increased power generation capacity.
- Extensive development and enhancement of electrical networks.
- Comprehensive planning for ancillary services to maintain system reliability and stability.

In this context, flexibility can play a major role in dealing with these challenges – This inclused also flexibility from other energy vectors including heat, gas, and transport. The interaction between different energy vectors could support the power system operation and result in a cost-effective decarbonisation of the energy system.

How can hydrogen facilitate the decarbonsination of heat (provide at least two applications)?

[2]

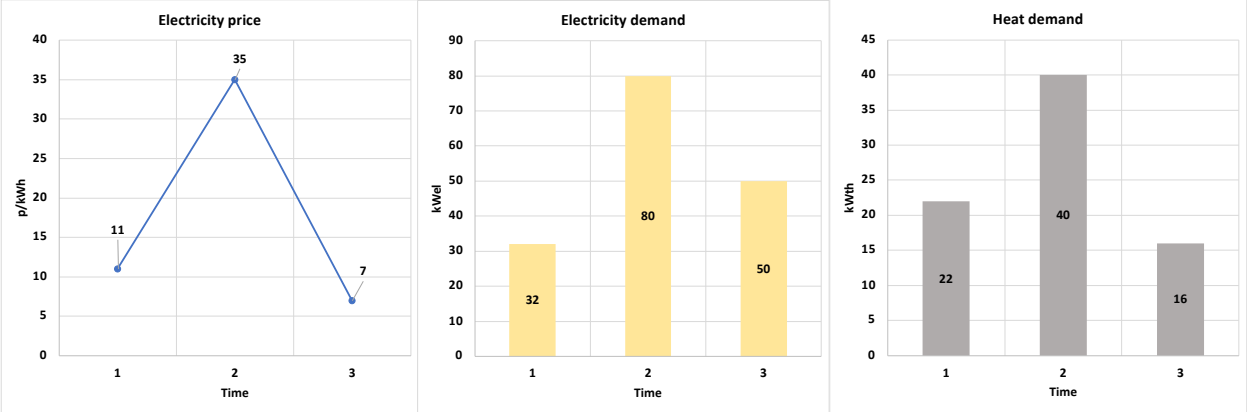
- Hydrogen can directly replace natural gas in parts of the distribution network, reducing disruption where electrification is difficult.
- Enables high-temperature industrial and district heating where heat pumps are less efficient.
- Facilitates seasonal energy storage, turning summer renewable excess into winter heat supply.
- Hydrogen-based CHP provides both heat and power, supporting grid balancing and local resilience.

Part II

Considering the electrification of the heat sector, heat appliances through thermal energy storage (TES) could provide flexibility to the electricity system and support consumer energy bills. Consider a system with a three-hour snapshot including the grid electricity price ∂_t^{Grid} and electricity demands of P_t^D and the heat demand of H_t^D.

- The employed CHP produces the same heat and electricity output, where the generation cost is ∂^{CHP} =8p/kW$_{\text{el}}$.
- The thermal energy storage (TES):
  - Maximum energy capacity of 10kWh.
  - TES is empty in t=0 and the energy at t=3 is 8kWh. Furthermore, it is assumed that at t=1, the TES will be charged for 6kWh.
  - Charging rate of 6kW and discharging rate of 4kW (i.e., in binary, meaning it is either 0 or 6kW (for charging) and 4 kW (for discharging)).
  - The heat loss rate is ignored, and the TES cannot be charged and discharged at the same time.

Model Answers



The objective is to minimise the consumers energy bills:

$$Obj = \min \sum_{t=1}^{3} \left(\partial_t^{\text{Grid}} \cdot P_t^{\text{Grid}} + \partial^{\text{CHP}} \cdot P_t^{\text{CHP}} \right)$$

The following equations on electricity and heat balances are provided:

$$P_t^D = P_t^{\text{CHP}} + P_t^{\text{grid}}, t = 1,2,3$$

$$H_t^D = H_t^{\text{CHP}} + H_t^{\text{TES,Dis}} - H_t^{\text{TES,Ch}}, \; t = 1,2,3$$

$$E_t^{\text{TES}} = E_{t-1}^{\text{TES}} + H_t^{\text{TES,Ch}} - H_t^{\text{TES,Dis}}, \; t = 2,3$$

a) Assuming that TES is not available, what is the total consumer energy bill during the provided time period?

[3]

As TES is not available, hence all the heat demand should be met by the CHP plants, therefore:

$$\partial_t^{grid} = (11, 35, 7); \; P_t^D = (32, 80, 50); \; H_t^D = (22,40,16)$$
$$H_t^{CHP} = H_t^D \Rightarrow H_t^{CHP} = (22,40,16) \Rightarrow P_t^{CHP} = (22,40,16)$$
$$P_t^D = P_t^{CHP} + P_t^{grid} \Rightarrow (32,80,50) = (22,40,16) + P_t^{grid} \Rightarrow P_t^{grid} = (10,40,34)$$

$$P_t^{CHP} = (22,40,16); \; P_t^{grid} = (10,40,34)$$

$$\text{Consumer bill} = \sum_{t=1}^{3} \left(\partial_t^{\text{grid}} \cdot P_t^{\text{grid}} + \partial^{\text{CHP}} \cdot P_t^{\text{CHP}} \right)$$
$$= 11 \times 10 + 35 \times 40 + 7 \times 34 + 8 \times (22 + 40 + 16) = 2{,}372 \, p = £23.72$$

Model Answers

b) Now, the TES can provide flexibility to the energy system, calculate the minimum consumer energy bill. Provide the full calculations.

[8]

Assumption: $E_0^{TES} = 0$; $H_1^{TES,Ch} = 6$

$t = 1$

$E_1^{TES} = E_0^{TES} + H_1^{TES,Ch} - H_1^{TES,Dis} = 6$

$\Rightarrow H_1^D = H_1^{CHP} + 0 - 6 \Rightarrow 22 = H_1^{CHP} - 6 = H_1^{CHP} = 28\ \text{kW}_{\text{th}}$

$\Rightarrow P_1^{CHP} = 28 \Rightarrow P_1^{grid} = 32 - 28 = 4\ \text{kW}_{\text{el}}$

For t=2 the TES can be either charged or discharged. Therefore, two possible combinations should be calculated to find the minimum consumer bill:

Mode I. Charging of TES:

$t = 2$

$H_2^{TES,Ch} = 6$; $H_2^{TES,Dis} = 0$

$E_2^{TES} = E_1^{TES} + 6 - 0 = 6 + 6 - 0 = 12$

Heat balance: $40 = H_2^{CHP} + 0 - 6 \Rightarrow H_2^{CHP} = 46$

$\Rightarrow P_2^{CHP} = 46 \Rightarrow P_2^{grid} = 80 - 46 = 34\ \text{kW}_{\text{el}}$

$t = 3$

As $E_3^{TES} = 8$, therefore the storage should be discharged for $4\ \text{kW}_{\text{th}}$:

$H_3^{TES,Dis} = 4$

Heat balance: $16 = H_3^{CHP} + 4 - 0 \Rightarrow H_3^{CHP} = 12$

$\Rightarrow P_3^{CHP} = 12 \Rightarrow P_3^{grid} = 50 - 12 = 38\ \text{kW}_{\text{el}}$

$P_t^{grid} = (4, 34, 38)$; $P_t^{CHP} = (28, 46, 12)$

$\Rightarrow \text{Consumer bill}_{\text{Mode I}} = 2{,}188\ p = £21.88$

Mode II. Discharging of TES : ($H_2^{TES,Dis} = 4$)

$t = 2$

$E_2^{TES} = 6 + 0 - 4 = 2$

$\Rightarrow 40 = H_2^{CHP} + 4 - 0 \Rightarrow H_2^{CHP} = 36$

$\Rightarrow P_2^{CHP} = 36 \Rightarrow P_2^{grid} = 80 - 36 = 44$

$t = 3$

As $E_3^{TES} = 8$, hence $H_3^{TES,Dis} = 0$ and $H_3^{TES,Ch} = 6$

$\Rightarrow 16 = H_3^{CHP} + 0 - 6 \Rightarrow H_3^{CHP} = 22$

$\Rightarrow P_3^{CHP} = 22 \Rightarrow P_3^{grid} = 50 - 22 = 28$

Model Answers

$P_t^{grid} = (4,44,28)\ ;\ P_t^{CHP} = (28,36,22)$

$\Rightarrow \text{Consumer bill}_{\text{Mode II}} = 2{,}486\ p = £24.86$

$\Rightarrow \text{Consumer bill} = \min(\text{Consumer bill}_{\text{Mode I}}, \text{Consumer bill}_{\text{Mode II}})$
$= \min(2{,}188 \text{ and } 2{,}486) = 2{,}188$

The minimum consumer bill will be achieved in the first mode, which will be 2,188p or £21.88.

c) Calculate the total savings in presence of TES on the consumer bill in %:

[1]

Savings associated to the case in part (a):

$$\text{Total savings} = \frac{2372 - 2188}{2372} \times 100 = 7.75\%$$

d) Assuming the electricity price grid at t=2 is unknown: $\partial_t^{\text{Grid}} = (11,X,7)$. What should be the price of electricity at t=2, so that charging/discharging of the TES does not impact the consumer bill.

[3]

For this it is required to make the grid price at time 2 variable and equal the consumer bill in both combinations presented in section (b):

$Mode1: P_t^{grid} = (4,34,38)\ ;\ P_t^{CHP} = (28,46,12)$
$Mode2: P_t^{grid} = (4,44,40)\ ;\ P_t^{CHP} = (28,36,10)$

$$11 \times 4 + \partial_2^{\text{Grid}} \times 34 + 7 \times 38 + 8 \times (28 + 46 + 12)$$
$$= 11 \times 4 + \partial_2^{\text{Grid}} \times 44 + 7 \times 40 + 8 \times (28 + 36 + 10)$$

$\Rightarrow \partial_2^{\text{Grid}} = 7\text{p/kWel}$

[20 marks]

Model Answers